FF-H-111C
May 15, 1974
SUPERSEDING
Int. Fed. Spec. FF-H-00111B(GSA-FSS)
June 10, 1960 and
Fed. Spec. FF-H-111A
July 8, 1948

FEDERAL SPECIFICATION

HARDWARE, BUILDERS'; SHELF AND MISCELLANEOUS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The hardware covered by this specification is that generally classified as miscellaneous builders' hardware. It is not intended for marine use (see 6.6).

1.2 Classification.

1.2.1 Types and sizes. The hardware furnished under this specification shall be of the types and, when applicable, sizes described in section 3, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of the invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-Z-363 - Zinc-base Alloy; Die Castings.
PPP-H-1581 - Hardware (Fasteners and Related Items) Packaging and Packing For Shipment and Storage of.

Federal Standards:

Fed. Test Method Std. No. 151 - Metals Test Methods.
Fed. Test Method Std. No. 406 - Plastics Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Markings for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

B287 - Acetic Acid-Salt Spray (Fog) Testing, Standard Method of.
B368 - Copper-Accelerated Acetic Acid-Salt Spray (Fog) Testing (CASS Test), Standard Method of Test for.
B487 - Measuring Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Standard Method for.
B499 - Measurement of Coating Thickness by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals, Standard Method for.
B504 - Measuring the Thickness of Metallic Coatings by the Coulometric Method, Standard Method for.
B530 - Measurement of Coating Thickness by the Magnetic Method: Electrodeposited Nickel Coatings on Magnetic and Nonmagnetic Substrates, Standard Method for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute Standard:

A156.6 - Architectural Door Trim.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 General.

3.1.1.1 Optional materials. Where designated materials are as listed below, the respective optional materials may be substituted. (This option does not apply to finishes.)

Designated material	Optional material
Brass and Bronze	Paragraph 3.1.1.2
Steel	Wrought iron
Cast nonferrous metal	Paragraph 3.1.1.3
Wrought (metal)	Paragraph 3.1.1.4

3.1.1.2 Brass and bronze. Where brass or bronze is specified herein as a material, without other requirements as to appearance or composition, the metals may be substituted interchangeably. Brass and bronze shall be suitable alloys composed chiefly of copper, with zinc and other elements as needed to provide the desired appearance and other properties usually associated with red bronze and yellow brass.

3.1.1.3 Cast nonferrous metal. Where cast nonferrous metal is specified, the supplier is allowed to substitute forged, wrought, or extruded forms of the same metal, provided the forged, wrought, or extruded items or parts are of substantially the same thickness or weight as commercially equivalent cast items or parts.

3.1.1.4 Wrought metal. Where wrought metal is specified, the rolled, forged, extruded, or pressure-cast form of the metal specified is acceptable.

3.1.1.5 Zinc-base alloy die castings. The castings shall conform to Federal Specification QQ-Z-363, and unless otherwise specified (see 6.2), may be of any composition specified therein.

3.1.1.6 Plastics.

3.1.1.6.1 General. Plastic materials used in hardware items shall have clean, smooth surfaces free from warpage and rough edges. Exposed surfaces shall be uniform in gloss. The part shall show no chipping or fracture where the "gate" is cut.

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3.1.1.6.2 Deformation in hot water. Plastics used in hardware items shall maintain their shapes without appreciable bending, warping, or contracting due to unmolding, when tested in accordance with 4.5.2.1.

3.1.1.6.3 Accelerated aging. Plastics shall not undergo appreciable changes in physical characteristics after being subjected to the accelerated aging treatment specified in 4.5.2.2.

3.1.1.6.4 Flammability. Plastics shall not burn at a rate greater than 3.0 inches per minute when tested in accordance with 4.5.2.3.

3.1.1.6.5 Resistance to discoloration. Plastic materials shall show no signs of streaking or discoloration when tested in accordance with 4.5.2.4.

3.2 Finishes.

3.2.1 General.

3.2.1.1 Standard finishes (see 6.4.3). Exposed surfaces of hardware shall have standard finishes as specified (see 6.2), in accordance with 6.3 and table I. If finish is not specified in bidding documents, items supplied shall be finished in accordance with 3.2.1.2.

3.2.1.2 Regular finishes. Where no other finishes are specified herein or in the bidding documents, brass shall have US3 finish, bright brass; bronze shall have US10, satin bronze; die cast zinc shall have US3 finish, bright brass; iron and steel shall be plated and have US10 finish; and aluminum shall have US23 finish, satin aluminum, anodized. When plain iron or steel is specified, it may be either finished or unfinished; a surface discoloration of rust will be acceptable unless otherwise specified (see 6.2).

3.2.1.3 Commercial finishes. For purposes of this specification a commercial finish shall be smooth, even, and bright, such as is produced by tumbling, burnishing, or chemically brightening, but not as highly reflective as produced by hand rubbing.

3.2.2 Lacquer. Unless otherwise specified (see 6.2), all natural color and plated hardware shall have a coating of transparent lacquer; except that chromium-plated, cadmium-plated, zinc-plated, and zinc-coated pieces may be furnished without lacquer.

3.2.3 Finishes on bronze. Finishes on bronze shall be uniform in texture with surfaces showing maximum smoothness and evenness obtainable by grinding, polishing, and buffing.

3.2.4 Finishes on steel, aluminum, and zinc alloy die castings. Plated finishes shall be polished or unpolished as specified (see 6.2), for the respective types. Bright-polished finishes shall have highly reflective surfaces showing maximum smoothness and uniformity such as is produced by grinding, polishing, and buffing. Dull polished surfaces shall be similarly produced and uniform in texture. Unpolished finishes shall be applied to surfaces that are suitably prepared by rolling or forging, with or without subsequent grinding or polishing, so as to produce a bright or satin finish, as required. These finishes are commonly referred to as "planished".

3.2.5 Chromium plating, finishes US 26 and US 26D. Chromium plating shall be applied over a coating of nickel as specified in 3.2.6. On significant surfaces, the minimum thickness of the chromium shall be 0.00001 inch when tested in accordance with 4.5.1.2, and chromium plating, applied over nickel plating on steel, shall show no appreciable corrosion on significant surfaces when tested in accordance with 4.5.1.6.

3.2.6 Nickel plating on iron, steel, aluminum, and zinc alloy die castings, US 14 finish. Nickel plating shall be applied directly to the base metal or over a coating of copper. On significant surfaces, the minimum thickness of nickel or of nickel and copper combined shall be 0.0004 inch when tested in accordance with 4.5.1.3 and 4.5.1.4, as applicable. If the nickel-copper combination is used, the minimum thickness of the nickel shall be 0.0002 inch when tested in accordance with 4.5.1.3 and 4.5.1.4, as applicable.

3.2.7 Aluminum finishes, US 23 and US 23A. The aluminum shall be treated so as to preserve a uniform surface, be free from breaks and scratches, and to show no evidence of aluminum corrosion when subjected to 125 hours of salt spray exposure in accordance with applicable portions of ASTM B287. In addition, the US 23 finish shall have an abrasion resistance equal to that obtained by a 0.0001 inch film of aluminum oxide such as is produced by the anodizing process.

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3.2.8 Cadmium plating, finish US 2C. The US 2C finish shall have a minimum thickness of 0.00015 inch on all significant surfaces when tested in accordance with 4.5.1.3, 4.5.1.4 and 4.5.1.5, as applicable.

3.2.9 Zinc coating, finishes US 2G and US 2H. Zinc coatings shall have a minimum thickness of 0.0005 inch on all significant surfaces when tested in accordance with 4.5.1.3, 4.5.1.4 and 4.5.1.5, as applicable. The coating shall be applied either by the hot-dip process or by electroplating, unless hot-dip, finish US 2H, is specified (see 6.2). The hot-dip process shall not be used to coat products which are fabricated of alloy or heat-treated steels which might be adversely affected by the heat employed in this process; in such cases electroplating shall be employed.

3.2.10 Primed for painting, finish USP. Hardware items which are primed for painting shall have an adherent paint coating of neutrol color that forms a suitable undercoat for subsequent finish painting. This coating shall be applied to iron or steel which has been suitably cleaned and bonderized.

3.3 Workmanship. Finished parts of hardware covered by this specification shall have true, accurately machined edges. Edges of unfinished steel and iron, and unpolished work need not be machine-finished but shall be smooth and reasonably true.

3.4 Fastenings. Each item of hardware shall be furnished complete with all necessary screws, bolts, and other fasteners required for proper installation. These fastenings shall be of suitable size and type, and shall harmonize with the hardware as to finish.

3.5 Illustrations. The illustrations appended hereto are descriptive of the desired types, and shall not be construed as showing required designs; however, items of designs other than those illustrated and described shall be full commercial equivalents of the ones shown. Numbers on each illustration correspond with numbers of the type, or the first of a group of types, to which it applies.

3.6 Tolerances.

3.6.1 Dimensions. Unless specifically identified herein, or unless, the sizes are delimited by the use of "maximum" or "minimum" dimensions, the dimensional tolerances for hardware items described in this document shall be ± 5%.

3.6.2 Weights. Unless specifically identified herein, or unless weight requirements are delimited by use of the term "maximum" or "minimum", the tolerances for weight and similar units of measure (force, pressure, etc.), relating to hardware items described in this document shall be ± 3%.

3.7 Individual types. Requirements for individual types shall be as follows:

3.7.1 Barrel bolts.

3.7.1.1 Regular duty.

Type 1011. - Wrought brass; minimum thickness of backplate 0.051 inch; lengths of bolt - 3, 4, and 5 inches.
Type F1011. - Wrought steel; japanned finish; otherwise, the same as type 1011.
Type F1011C. - Wrought steel; zinc coated; otherwise, the same as type 1011.

3.7.1.2 Heavy duty.

Type 1012. - Wrought brass; minimum thickness of backplate 0.065 inch; length of bolt - 4, 6 and 8 inches.
Type F1012. - Wrought steel; japanned finish; otherwise, the same as type 1012.
Type F1012C. - Wrought steel; zinc coated; otherwise, the same as type 1012.
Type 1013. - Cast bronze; lengths of bolt - 4 and 5 inches.

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3.7.2 Chain and foot bolts.

Type 1021A. - Chain bolt. Spring-action; cast bronze; lengths of case, 4 and 6 inches; regular and angle (reverse bevel) strikes.
Type F1021A. - Chain bolt. Spring-action; cast iron; otherwise the same as type 1021A.
Type 1021B - Foot bolt, cast bronze; lengths of case, 4 inches and 6 inches.
Type F1021B. - Foot bolt. Cast iron, otherwise the same as type 1021B.
Type 1023A. - Chain bolt. Spring action; wrought brass; round or square case regular or angle (reverse bevel) strikes; lengths of bolt, 3 inches and 6 inches; minimum thickness of plate, 0.040 inch for 3 inch bolt and 0.064 inch for 6 inch bolt.
Type F1023A. - Chain bolt. Spring action; wrought steel; japanned finished, round or square case; regular and angle (reverse bevel) strikes; bolt lengths, 3, 6, 8 and 10 inches; minimum thickness of plate, 0.040 - 3 inch, 0.064 - 6 inch, 0.064 - 8 inch, 0.072 - 10 inch.
Type F1023AC. - Chain bolt. Wrought steel, zinc plated, otherwise the same as type F1023A.
Type 1023B. - Foot bolt. Wrought brass, round or square case; lengths of bolts, 3 inches and 6 inches; minimum thickness (inches) of plate, 0.040 for 3 inch and 0.064 for 6 inch.
Type F1023B. - Foot bolt. Wrought steel; japanned finish; round or square case; lengths of bolts, 3, 6, 8 and 10 inches; minimum thickness (inches) of plate, 0.040 - 3 inch, 0.064 - 6 inch, 0.064 - 8 inch, 0.072 - 10 inch.
Type F1023BC. - Foot bolt. Wrought steel, zinc plated; otherwise the same as F1023B.

3.7.3 Lever extension flush bolts.

Type 1045. - Cast brass or aluminum, plate and strike; plate 6-3/8" x 1-1/4"; bolt head 1/2" square, brass, aluminum, or nylon; length of rod 12".
Type 1045R. - For rabbeted doors. Otherwise same as type 1045.
Type 1046. - Cast bronze plate, 5-7/8" x 3/4". Bolt head to be of brass, aluminum, or nylon, 3/8" diameter. Length of rod 12". Suitable for 1-3/8" doors. Strike to be of brass or aluminum.
Type 1048. - Cast bronze dust-proof strike to suit type of bolt specified (see 6.2). Approximately 1" x 2-1/2" on face by 2" depth; back fully enclosed.
Type 1049. - Cast brass or aluminum front plate and strike. Front plate 6-3/4" x 1" x 1/8". Strike 2-1/4" x 15/16". Backset 3/4". Length of rod 12". (This item shall be suitable for installation on hollow metal doors and frames prepared in accordance with ANSI Standard A115.4.)
Type 1050AB. - Automatic flush bolt. Wrought brass. Designed exclusively for use on metal doors swinging in pairs. To be installed top and bottom on the inactive leaf. Bolts are activated by closing of the active leaf and released automatically by opening the active leaf. Shaft is to be nickel plated. Bolt shall fit 1-3/8 inch to 2-1/4 inch doors. Finish of exposed parts shall be bright brass.
Type 1050AS. - Same as type 1050AB, except material shall be stainless steel with satin finish.

3.7.4 Cane bolt.

Type F1051. - Heavy wrought steel; zinc-plated rod with minimum diameter of 5/8 inch; strikes for wood and for concrete; suitable stops shall prevent the bolt from slipping out of the guides; minimum throw 6-1/2 inches; lengths of bolt, 18 inches and 24 inches.

3.7.5 Mortise door bolt.

Type 1053. - Wrought bronze with steel or nylon bolt. Diameter of case approximately 9/16"; backset minimum 1-5/8".

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3.7.6 Surface bolts.

Type 1061. - Wrought bronze; 1/2 inch half-round or oval rod; 7/8 inch throw; two guides which shall prevent the bolt from touching woodwork. Bolt shall not rotate and shall be secure in both extended and withdrawn positions. Furnish with angle and universal strikes; lengths of 6 inches and 9 inches.
Type F1061 - Wrought steel; otherwise same as type 1061.
Type 1062. - Wrought bronze; 3/8 inch half-round or oval rod; lengths of 4 inches and 6 inches; otherwise same as type 1061.
Type F1062. - Wrought steel; otherwise same as type 1062.

3.7.7 Shelf rests.

Type F1066. - Wrought steel, nickel-plated. Length of pin, 3/8"; diameter, 1/4". overall length approximately 1-1/4".
Type F1066A. - Same as type F1066, except cadmium plated.

3.7.8 Adjustable shelf supports.

Type 1067. - For surface application. Wrought brass, nickel-plated. Screw fastenings for standard spaced not over 6 inches, rests shall hook securely into slots in standard. Shall provide adjustment of shelf by 1/2" increments. Alternate slots shall be numbered in groups of not less than 6. Minimum projection of rest, 3/4"; minimum width, 9/16". Minimum thickness of standard and rest, 0.048". Unless otherwise specified (see 6.2), furnish standard in 12 foot lengths, with one style "A" rest per foot.
Type F1067. - For surface application. Wrought steel, nickel-plated. Otherwise the same as type 1067.
Type 1067G. - For flush application in groove. Wrought brass. Same as type 1067, except formed for flush mounting.
ype F1067G. - For flush application in grove. Wrought steel. Same as type 1067, except formed for flush mounting.

3.7.9 Shelf brackets.

3.7.9.1 Fixed brackets.

Type F1068. - Wrought steel, japanned or enameled finish at the manufacturer's option; one-piece tapered rib construction. Dimensions of brackets and fasteners shall be as follows:

TABLE I. Dimensions and screws for type F1068

Size of bracket Inches	Thickness of plate, minimum Inch	Shelf plate screws, minimum			Wall plate screws minimum		
		No.	Length Inch	Size No.	No.	Length Inches	Size No.
4 by 5	0.035	3	3/4	6	3	1-1/4	6
5 by 6	.035	3	3/4	6	3	1-1/4	6
6 by 8	.047	3	3/4	6	3	1-1/4	6
8 by 10	.062	3	3/4	7	3	1-1/4	7
10 by 12	.062	3	3/4	7	3	1-1/4	7
12 by 14	.068	3	3/4	8	3	1-1/4	8
*16 by 18	.095	4	1	10	4	1-1/2	10

*Equipped with center brace.

3.7.9.2 Combination bracket with closet pole.

Type 1068C. - Wrought steel, size 9-3/4 inches high by 11 inches wide, minimum thickness 0.102 inch. Shall support a combined load of 125 pounds on the rod support section of the bracket and 150 pounds on the shelf support section 8 inches from the wall directly over the bracket, with a deflection of not more than 1/8 inch, when properly mounted.

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3.7.9.3 Adjustable brackets with standards.

Type 1068A. - Wrought steel, minimum thickness .057 inch; available sizes 4, 5, 6, 8, 10, 12, 14, 16, 18, and 20 inches. Standards shall be 5/8 inch wide and 7/16 inch deep, lengths 12, 18, 24, 30, 36, 42, 48, 60, 72, 84, 96, and 144 inches. One bracket shall be included for each 12 inch length of standard, unless otherwise specified (see 6.2).

Type 1068AA. - Aluminum, minimum thickness .079 inch; otherwise same as type 1068A.

3.7.10 Closet and cabinet latches.

3.7.10.1 Friction catches.

Type 1070. - Wrought brass, size 2 inches x 3/4 inch. Spring action plunger. For mortise application.

Type 1071. - Wrought brass, tubular housing, 3/8 inch diameter x 1/2 inch deep. Spring action polished steel rounded plunger or ball. Rounded concave strike of wrought steel, 5/8 inch diameter. Plunger and strike to be plated. For application in bored holes.

Type F1072. - Tempered steel, one-piece spring clip, 1/2 inch to 1 inch x 1 3/8 inch; having two prongs engaging wrought steel strike of 1/2 inch x 1 3/8 inches. For surface mounting. Zinc plated, except spring may have blued finish.

Type F1073. - Steel. Housing 1/2 inch x 1/2 inch with base 1 1/4 inches x 3/4 inch. Elongated holes for surface mounting. Two floating metal prongs held by spring tension to engage strike mounted on door. Strike 3/8 inch x 1 1/8 inches. Zinc plated.

Type F1073D. - Wrought steel with rubber rollers. Housing 1/2 inch x 3/4 inch with base 1 7/16 inches x 3/4 inch. Elongated holes for surface mounting. Two rubber rollers held by spring tension to engage strike mounted on door. Strike 3/8 inch x 1 1/8 inch, zinc-plated

Type F1073S. - Wrought steel case and strike. "Spring cushioned" rubber rollers. Housing size 1 1/8 inches x 1 3/8 inches, strike 2 1/16 inches x 5/8 inch and 1 5/16 x 1 inch. Elongated screw holes for surface mounting. Zinc-plated.

3.7.10.2 Elbow catches.

Type 1075. - Cast brass, unpolished. Spring action catch. Size 5/8 inch x 1 1/4 inches.

Type F1075. - Cast iron, unpolished. Spring action catch. Size 5/8 inch x 1 1/4 inches.

Type F1075A. - Wrought steel, nickel-plated. Spring action catch. Size 1 inch x 1 5/16 inches.

3.7.10.3 Magnetic catches.

Type 1079. - Aluminum or plastic case, 2" x 1-1/4". Ceramic magnet. Steel strike, zinc-plated 1-1/2" x 11/16". Elongated screw holes on case. "Floating magnet" to compensate for warping of shrinking of door. Minimum pull 8 pounds.

3.7.10.4 Cupboard turns.

Type 1083. - Cast bronze. Oval knob. Size, without strike 2 1/4 inches x 1 1/2 inches. Latch bolt, minimum 15/32 inches x 7/16 inches; minimum throw 5/16 inch.

Type 1085. - Wrought bronze. Minimum thickness of case 0.040 inches. Otherwise the same as type 1083.

Type F1085. - Wrought steel. Minimum thickness of case 0.040 inches. Otherwise the same as type 1083.

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3.7.11 Screen door catches.

3.7.11.1 Push-pull type.

Type 1086. - Brass aluminum. For screen, storm or combination doors from 3/4 inch to 1 3/8 inches thick. Surface applied by boring a 1 inch hole in door stile. Equipped with safety locking feature operated from inside only. Handle 2 1/2 inches long.

Type F1086. - Wrought steel or zinc die cast. Otherwise the same as type 1086.

3.7.11.2 Rim type.

Type 1089. - Cast or wrought bronze case, 2 1/4 inches x 1 7/8 inches. Wrought bronze outside escutcheon with beveled or rolled edges, and knob; bronze lever. Stop in case to dead-lock latch bolt. Bolt throw 1/2 inch. For doors 7/8 inch to 1 1/4 inches thick, opening out.

3.7.11.3 Bored type.

Type 1090. - Tubular case-minimum 9/16 inch diameter. Brass or wrought steel. Bolt-brass, 1/2 inch minimum diameter. Backset 1 1/8 inches minimum. Strike-bronze, with lip. Knob-minimum 1 3/4 inches diameter. Wrought bronze. Lever handle-minimum length 1 5/8 inches cast or bar bronze. Escutcheons: If rectangular, minimum 4 inches x 1 3/8 inches. If round, minimum diameter 2 inches. Wrought bronze. Operation: By knob or lever handle, except when locked from inside. Reversible.

Type 1090S. - For doors 3/4 inch to 1 1/8 inches. Material: Brass or aluminum. Knob: 1 3/4 inches minimum. Lever handle: 1 11/16 inches minimum. Roses: Oval or round 1 3/4 inches minimum. Operation: By knob outside or lever inside, except when knob or lever have been locked inside.

3.7.12 Casement sash and cupboard top closer.

Type F1133. - Wrought steel. Zinc plated. Spring actuated lever pivoted in formed case and arranged to engage hook sash bracket on sash when not less than 1" from closed position. On opening of sash, shall move to an remain in position for engagement on closing the sash. Minimum width of case 1"; minimum length 3-3/16".

3.7.13 Casement fasteners.

Type 1135. - Cast bronze lever handle on 2 inches x 1 inch minimum size plate. Projection 1 1/2 inches minimum. Mortise, rim or surface strike to suit application.

Type F1135. - Cast iron, wrought steel or zinc alloy die castings. Otherwise the same as type 1135.

Type 1135AL. - Aluminum. Otherwise the same as type 1135.

Type 1136. - Cast or wrought bronze. Lever handle on 2 inches x 1 inch minimum size plate. Projection 1 inch. Mortise, rim, or surface strike to suit application.

Type F1136. - Cast iron, wrought steel or zinc alloy die casting. Otherwise the same as type 1136.

3.7.14 Door holders.

3.7.14.1 Nonautomatic.

3.7.14.1.1 Lever types.

Type 1149A. - Cast bronze. Replaceable rubber tip. Lever shall swing up and remain securely held in vertical position when out of action. Length 4 inches to 4 1/2 inches for doors with 2 inches or less floor clearance.

Type 1149B. - Same as type 1149A, except with lever 5 inches to 5 1/2 inches long for doors with 3 1/2 inches or less floor clearance.

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3.7.14.1.2 Plunger types.

Type 1154. - Cast bronze case, steel catch plate and steel or malleable iron plunger rod with rubber tip. Steel parts plated. Rubber tip replaceable. Releasing lip shall be plainly evident. Size 5 3/4 inches (minimum height). Minimum throw of plunger 1 inch.

Type F1154. - Cast iron case. Size 5 3/4 inches. Otherwise the same as type 1154.

Type 1156. - Wrought bronze case. Size 5 3/4 inches. Otherwise the same as type 1154.

Type F1156. - Wrought steel case. Size 5 3/4 inches. Otherwise the same as type 1154.

Type 1158. - Cast bronze case; steel catch plate and steel plunger rod with rubber shoe. Steel parts plated. Rubber shoe flanged and held in place by knurled and threaded bronze collar so as to be replaceable without removing holder from door. Size not less than 6 3/4 inches x 1 5/16 inches over-all. (Two acceptable designs are illustrated.) Releasing lip shall be plainly evident.

3.7.14.2 Automatic.

3.7.14.2.1 Concealed overhead slide.

Type 1160. - For single-acting entrance and vestibule doors hung on butts and opening to 110°. Brass or bronze. Extruded channel having a minimum nominal perimeter of 4 inches and a minimum wall thickness of 1/8 inch fastening to door at each end and at intermediate points as required by the length of the channel. Arm and bracket 3/16 inch thick, minimum, width to suit thickness of door, but not less than 1 1/4 inches. Stop with encase shock absorber effective at all times. Hold open and release by push and pull on door, except when exposed control knob is set in inactive position. Holding pressure adjustable for various conditions. Applied in mortise in top rail of door; arm exposed when door is open. Size to suit width, thickness, and swing of door.

Type 1160A. - Same as type 1160, except for doors hung on offset pivots.

Type 1160D. - For double-acting entrance and vestibule doors. Same as type 1160, except width of arm same as thickness of door.

3.7.14.2.2 Surface-mounted overhead slide.

Type 1161. - For entrance and vestibule doors hung on butts and opening to 95° unless other-openings from 90° to 110° are specified (see 6.2). Brass or bronze. Extruded track or channel. Arm and bracket 1/4 inch thick, minimum. Stop with encased shock absorber effective at all times. Hold-open and release by push and pull on door, except when control knob is set in inactive position. Holding pressure adjustable for various conditions. Applied to surface of top rail of door with 1/4 inch or larger machine screws and sleeve bolts through door. Size to suit width, thickness, and swing of door.

Type 1161A. - Same as type 1161, except for doors hung on offset pivots.

3.7.14.2.3 Surface-mounted, with overhead joint.

Type 1162. - For entrance and vestibule doors opening to 110°. Cast phosphor bronze or gun metal. Disks from joint providing larger horizontal bearing area, and having stop lugs. Encased shock absorber with compression spring. Hold-open and release by manual operation of holding catch. Stop and shock absorber effective at all times. Furnish 15 inch extension handle when specified (see 6.2).

Size A. - For doors 24 inches to 29 inches wide. Minimum weight, 3 lbs.

Size B. - For doors 30 inches to 35 inches wide. Minimum weight, 3 lbs., 10 oz.

Size C. - For doors over 35 inches wide. Minimum weight 3 lbs., 14 oz.

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Type F1162. - Same as type 1162, except with malleable iron arms, plated; bronze bearings and fittings.

3.7.14.2.4 Concealed overhead friction slide.

Type 1164. - For interior doors opening to 95° unless other openings from 90° to 110° are specified (see 6.2). Brass or bronze. Extruded channel 5/8 inch x 11/16 inch x 1/16 inch thick, minimum. Arm and bracket 3/16 inch thick, minimum. Frictional element made of asbestos non-abrasive brake lining having a minimum effective face area of 1 1/4 sq. inches, and held under adjustable pressure by meatal spring. Free-acting shouldered pivots. Shock absorber in one end of channel. Applied in mortise in top rail of door. Size to suit width, thickness, and swing of door. Shall hold in all positions.

3.7.14.2.5 Surface-mounted overhead friction slide.

Type 1165. - For interior doors opening to 110°. Same as type 1164 except for application to surface of top rail of door by machine screws and sleeve bolts through door.

3.7.14.2.6 Surface-mounted overhead rod.

Type 1165R. - For single acting entrance and vestibule doors hung on butts, offset pivots, or center hung pivots, and opening to 110°. Round steel rod not less than 1/2 inch diameter with shock absorbing spring. Bronze jam bracket, pivot, and door bracket. Steel or nylon roller. Hold open and release by push and pull on door except when roller cam is set in inactive position. Adjustable spring tension. Stop and shock absorber effective at all times.

Size A - For door widths up to 28 inches.
Size B - For door widths between 29 and 34 inches.
Size C - For ddor widths between 35 and 40 inches.
Size D - For door widths between 41 and 46 inches.
Size E - For door widths between 47 and 53 inches.

3.7.14.2.7 Concealed overhead slide.

Type 1166. - For interior doors of medium size, single or double acting, opening to 95 degrees unless other openings from 90 degrees to 110 degrees are specified. Bronze or steel channel and arm; channel 0.050" thick, minimum; arm and bracket 3/16" thick, minimum; widths to suit thickness of door. Stop with shock absorber. Hold-open and release by push and pull on door. Applied in mortise in top rail of the door; arm exposed when door is open. Size to be determined by the width, thickness, and swing of the door.

Type 1166A. - Stay for interior doors of medium size. Same as type 1166, except without hold-open.

3.7.14.2.8 Floor types.

Type 1167. - For entrance and vestibule doors. Cast bronze. Encased spring bumper with metal plunger and stop. Door strike plate with projecting hook to engage hinged holding tongue on top of bumper. Operation by push on door to stop position, when engagement is automatic; release by quick push or door. Holding tongue inactive when placed in reverse position. Height 3-1/2" for doors with 1-1/2" maximum floor clearance, unless otherwise specified (see 6.2). Base shall be at least 3-1/2" x 2". Attachment to floor by a 1/2" steel stud with expansion sleeve and a 1/4" - #20 machine screw with expansion sleeve.

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Type 1168. - Cast bronze. Push and pull. Spring acturated engaging roller enclosed in housing attached to door by four screws. Strike attached to floor by three screws. Screws shall be wood or metal screws, or machine screws with expansion sleeves, as specified (see 6.2). Rubber bumper. Housing at least 3-1/4" x 2" at base; 2" maximum projection. Strike 2" x 2-1/4" minimum at base with height to suit clearance between door and floor.

Type 1168A. - Same as type 1168 except with independently adjustable tension and projection spring actuated nylon roller latch encased in housing approximately 3" x 1-3/4"; overall height not over 4"; attached by four concealed screws; with adjustable rounded dome turret strike, changeable from holder position to a cusion stop, with base approximately 3" in diameter, full round or cutaway.

3.7.15 Closet hanger bars.

Type 1171. - Wrought brass telescoping tubing, nickel plated. Minimum wall thickness 1/32". Seamless outer tube not less than 1" outside diameter. A metal wall plate shall have at least 2 screw holes for attaching to hook strip in closet. The tube shall be securely attached or locked to the wall plate when installed. Size 18" shall be adjustable from 18" to 30"; size 30" shall be adjustable from 30" to 48"; and size 48" shall be adjustable from 48" to 72". Furnish center support with 48" size.

3.7.16 Clothing hooks.

3.7.16.1 Coat and hat hooks.

Type 1172. - Cast bronze, upper prong bent. Two screws. Projection 3 inches. Commercial finish.

Type F1172. - Cast iron or zinc alloy die casting. Otherwise the same as type 1172.

Type 1172AL. - Aluminum. Otherwise the same as type 1172.

Type F1173. - Steel wire. Diameter 0.135 inch (No. 10 gage). Screw point projection 3 inches.

3.7.16.2 Robe hooks, bathroom.

Type 1174. - Brass, chromium plated. Concealed wall fastenings; two hooks may be integral with or permanently fastened to wall flange. Projection of 2 inches from back of flange to ends of hooks.

Type 1175. - Zinc alloy, chromium finish, concealed wall fastening, two hooks integral with the wall flange. Projection of 1 5/8 inches from back of flange to end of hooks.

3.7.17 Sash and screen lifts.

3.7.17.1 Bar types.

Type 1203. - Cast bronze. Heavy, rounded corners, 1 3/8 inches x 5 inches on bases.

Type 1204. - Cast bronze. Light, rounded corners, 1 1/16 inches x 4 inches on bases.

Type F1204. - Cast iron or zinc alloy die casting. Otherwise the same as type 1204.

Type 1204AL. - Aluminum. Otherwise the same as type 1204.

Type 1206. - Pressed steel lift brass plated or pressed solid brass, polished. Bearing on bottom plate to be 4 inches and furnished with three #7 x 3/4 inch oval head brass plated screws.

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3.7.17.2 Flush types.

Type 1212. - Cast bronze. Square corners. Size 1 3/4 inches x 3 inches minimum. Side screw holes. Lifting surfaces shall be smooth and free from sharp, rough edges.
Type 1213. - Wrought bronze. Otherwise the same as type 1212.
Type F1213. - Wrought steel. Otherwise the same as type 1212.
Type 1213AL. - Aluminum. Otherwise the same as type 1213.

3.7.17.3 Screen lifts.

Type 1223. - Wrought bronze. Length 3 1/2 inches.
Type F1223C. - Wrought steel, zinc-plated. Length 3 1/2 inches.

3.7.18 Door hardware.

3.7.18.1 Door protection plates. Material, workmanship, and tolerance requirements shall be in accordance with requirements stated for these items in American National Standard A156.6; Product type J100.

Type 1224. - Wrought brass. Edges not beveled unless otherwise specified (see 6.2). Height and width as specified (see 6.2).
Type 1225. - Stainless steel. US32D finish. Edges not beveled unless otherwise specified (see 6.2). Height and width as specified (see 6.2).
Type 1226. - Wrought aluminum. US28 finish. Edges not beveled unless otherwise specified (see 6.2). Height and width as specified (see 6.2).
Type 1226A. - Same as type 1226, except with bright finish.
Type 1227. - Plastic plate. Black color to be furnished unless chocolate brown is specified (see 6.2). Uniform satin matte finish on face. Height and width as specified (see 6.2).

(a) Material. - The plastic material shall be a laminate, molded in sheet form with suitable base or filler properly impregnated and bonded with a thermosetting resin compound or binder processed to make it hard, non-absorbent and free from porosity and brittleness, so as not to chip or crack in normal service, and so as to meet the following requirements in addition to the requirements of 3.1.1.6.2, 3.1.1.6.3, 3.1.1.6.4, and 3.1.1.6.5.
(b) Impact strength. - The plates shall have an impact strength of not less than 0.50 foot-pounds per inch of notch at 25°C. (77°F.) when tested as specified in 4.5.2.5.
(c) Resistance to boiling water. - The increase in weight after boiling in water for 2 hours shall not exceed 3 percent of the original weight, and the increase in thickness of the same samples shall not exceed 5 percent of the original thickness when tested as specified in 4.5.2.6. Visual examination of samples after the boiling test shall reveal no delamination, cracking, or splitting and no blistering, crazing, or checking of the surfaces.
(d) Abrasion resistance. - In the abrasion test specified in 4.5.2.7, the average loss in weight shall not exceed 30 milligrams per 1000 revolutions.
(e) Resistance to chemicals. - The face of the plates shall show no spotting or blistering after 24 hours exposure to chemicals as specified in 4.5.2.8.

3.7.18.2 Door pulls. Material, workmanship, and tolerance requirements shall be in accordance with requirements stated for these item in American National Standard A156.6; Product type J400.

Type P1228. - Cast brass, plain design. Grip is 5-1/2 inches center to center with integral round bases 2 inches in diameter.
Type P1228P. - Cast brass, plain design, on plate. Minimum size of plate 14 inches x 3 inches; grip shall be 5 inches center to center.

3.7.18.3 Push plates. Material, workmanship, and tolerance requirements shall be in accordance with requirements stated for this item in ANSI Standard A156.6; Product type J300.

Type PP1228W. - Wrought brass, plain design plate. Size 14 inches x 3 inches.

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3.7.18.4 Push-pull bars. Material, workmanship, and tolerance requirements shall be in accordance with requirements stated for these items in ANSI Standard A156.6; Product type J500.

Type B1228. - Push bar, plain design. Cast brass bases, 2 inches in diameter. Tubing shall be wrought brass of 1 inch diameter. Projection of the bar shall be 2-1/2 inches. Length of the bar shall 24, 30, or 36 inches, as specified (see 6.2).

Type B1228S. - Extruded brass, solid bar; 1-1/4 inches x 3/8 inch in cross section. Furnished with or without bases, as specified (see 6.2). Bases to be 3/8 inch thick. Bar length shall be 24, 30, or 36 inches, as specified (see 6.2).

Type B1228PP. - Cast brass, plain design. Two type B1228 bars with 8 inch connecting pull. Fastened by machine screws through door.

Type B1228PPS.- Two type B1228S bars, 8 inches center to center with connecting pull. Fastened by machine screws through door.

Type B1228SD. - Screen door guard. Wrought brass, U-shaped construction. Width 1 inch, projection 3/4 inch, thickness of metal 0.058 inch at ends and 0.049 inch in middle section. Adjustable for doors 30 inches to 42 inches wide.

3.7.19 Sash pulleys.

Type 1230. - Turned brass wheel, with combination groove for cord or chain. Minimum width of groove 0.360 inch. Front not less than 5 inches x 1 1/8 inches x 1/8 inch, width wrought bronze face plate. Cast iron case with round ends. Roller bearing with 1/4 inch minimum diameter steel axle and rollers machined true. Wheel diameter 2 1/4 inches, unless 2 1/2 inches diameter wheel is specified (see 6.2).

Type 1230A. - Turned iron wheel. Otherwise the same as type 1230.

Type 1230B. - Same as type 1230, except with turned iron wheel and plain axle. If wood bushing is used, it shall be of hardwood impregnated so as to withstand the test in 4.5.3.

Type SA1231. - (For special application, see 6.3). Overhead type sash pulley. Turned-face iron wheel. Wrought bronze front. Cast-iron case. Roller bearings, rollers machined true. Wheels fully housed to prevent chain from leaving wheel. Equipment shall include all necessary parts of suitable sizes for complete operation on double-hung counter-weighted sash in mullion windows without weights in the mullions.

Type SA1231A. - Same as type SA1231, except with plain axle. If wood bushing is used, it shall be of hardwood impregnated so as to withstand the test in 4.5.3.

Type 1236. - For sliding sash or grille. Cast brass or bronze case and wheel; steel axle. 3 1/4 inches x 1/2 inch front; 1 inch diameter wheel grooved for 1/8 inch diameter wire cord. Wheel with steel bushing and antifriction bearings.

Type 1236A. - Cord attachments for use with type 1236 pulley. Lug for sash end, both soldered. Shall be suitable for type 1236C wire cord.

Type 1236C. - Wire cord, for use with type 1236 pulley. Phosphor bronze. Six strands having 19 wires each. Cotton core. Overall diameter 1/8 inch. Tensile strength 400 pounds minimum. Length to suit application.

3.7.20 Sash pole hooks.

Type 1264. - Cast bronze. Length 3-3/4". Diameter 1-1/8".

3.7.21 Sash poles.

Type 1265A. - Bronze hook on handle of japanned or enameled steel tubing. Minimum wall thickness of tubing 0.035 inch; diameter 3/4 inch ± 1/16 inch. Lower end of handle protected with rubber cap having steel washer insert. Length 6 feet, unless 8- or 10-foot lengths are specified (see 6.2).

Type 1265B. - Hardwood. 1-1/8 inches minimum diameter. Chucked to fit sash pole hook. Length 6 feet long, unless otherwise specified (see 6.2). Poles exceeding 12 feet in length may be made in two sections with brass screw joint coupling.

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Type 1265AL. - Cast aluminum hook attached to pole of seamless aluminum tubing of 7/8 inches outside diameter and .0625 inch minimum wall thickness. Lower end of pole protected by rubber tip cemented to it. Length 6 feet, unless 8- or 10-foot lengths are specified (see 6.2).

3.7.22 Sash pull plates.

Type 1266. - Cast bronze. Minimum size 1" x 2".

3.7.23 Pulls.

3.7.23.1 Screen door pulls.

3.7.23.1.1 Light pull.

Type 1269. - Wrought bronze. (Par. 3.6.3). Rounded grip closed in back. Length 4 3/4 inches.
Type F1269. - Wrought steel. Otherwise the same as type 1269.

3.7.23.1.2 Heavy pull.

Type 1274. - Cast bronze. Length 6 inches. Four screw holes. Grip of flat oval section, with all exposed surfaces polished. Minimum weight 4 1/4 pounds per dozen without screws.
Type 1275. - Wrought bronze. Minimum length 6 inches. Flat oval grip closed in back. Four screw holes.
Type F1275C. - Wrought steel, zinc-plated. Otherwise the same as type 1275.
Type F1276. - Wrought steel, extra heavy. Japanned finish. Length approximately 9 1/2 inches over-all. Six screw holes.

3.7.23.2 Knob pulls.

Type 1280. - Cast bronze round knob and shank. Attachment by steel washer-head machine screws or truss head machine screws, zinc-plated.

TABLE II. Dimensions of knob pulls

Sizes, inches	1	1-1/8	1-1/4	1-1/2
Diameter of Knob, inches	1	1-1/16	1-1/4	1-1/2
Projection, Min., inches	1	1	1-1/8	1-1/4

Type 1280D. - Same as type 1280 except of zinc die-cast alloy. Chromium-plated.
Type 1281. - Seamless wrought bronze top; solid bronze shank. Otherwise the same as type 1280.
Type 1282. - Wrought brass knob and shank. Pattern optional. Otherwise the same as type 1280.
Type 1285. - Zinc alloy die cast knob and shank. Chromium plated. Attachment by steel trimit or washer head machine screws, zinc-plated. 1-1/16 inches diameter.
Type 1286. - Machined brass knob and shank. Attachment by steel trimit or washer head machine screws, zinc-plated. 1-1/16 inches, 1 1/4 inches, and 1-1/2 inches diameter.
Type 1286AL. - Aluminum anodized. Otherwise the same as type 1286.
Type 1287. - Wrought brass knob and shank. Pattern optional. Attachment by steel trimit or washer head screw, zinc-plated.

3.7.23.3 Drawer pulls.

Type 1305. - Cast bronze. Rounded handle; rectangular bases. Size, 4 inches in over-all length. Two steel washer-head machine screws, zinc-plated for attachment from inside of drawer.
Type 1306. - Cast bronze. Modern design, pattern optional. Size 2-3/4 inches to 5 inches in over-all length. Two steel washer-head or trimit machine screws, zinc-plated for attachment from inside of drawer.
Type 1306D. - Same as type 1306 except of die-cast zinc alloy, chromium plated.
Type 1307. - Wrought brass. Grip with closed back, flat or rounded. Size, 3-3/8 inches to 4-3/4 inches in over-all length. Otherwise the same as type 1306.

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Type F1307. - Wrought steel. Chromium-plated. Grip with flat back. Size 3-3/8 inches to 4-3/4 inches in over-all length. Otherwise the same as type 1306.
Type 1308. - Wrought brass, minimum thickness 0.040 inch. Size 3-1/2 inches to 4-1/2 inches x 1/2 inch to 5/8 inch over all. Two oval-head wood screws for attachment from front.
Type F1308. - Wrought steel, chromium-plated. Otherwise the same as type 1308.
Type 1310. - Wrought bronze. Cup pattern. Minimum length 3-1/4 inches.
Type F1310. - Wrought steel, chromium-plated. Cup pattern. Length not less than 3-1/4 inches.

3.7.24 Chain door stops.

Type F1317. - For screen and storm doors opening out. Steel chain of flat link type in one length; compression spring or springs minimum 4-1/4 inches in total length and 3/4 inch in outside diameter attached to one or both brackets and shain. Spring shall withstand at least 100 pounds pull on chain before reaching full compression. Tensile strength of chain and attachments minimum 350 pounds. Brackets minimum 0.072 inch thick. Over-all length minimum 22 inches. Zinc-plated.

3.7.25 Door stops.

Type 1320. - Cast bronze, heavy weight. Rubber tip. Projection 3-1/2 inches. Diameter of base 2-1/4 inches. Three wood screws.
Type 1320E. - Same as type 1320, except for attachment by steel stud with expansion shield and auxiliary machine screw with expansion shield.
Type 1321. - Same as type 1320, except with hook and keeper.
Type 1321E. - Same as type 1320, except with hook and keeper, and attachment by steel stud, with expansion shield and auxiliary machine screw with expansion shield.
Type 1322. - Cast bronze, medium weight. Rubber tip. Base with wood screw. Projection 3 inches. Diameter of base 1 inch.
Type F1322. - Cast iron or zinc alloy die casting. Otherwise the same as type 1322.
Type 1322AL. - Aluminum. Otherwise the same as type 1322.
Type 1323. - Cast bronze, light weight. Rubber tip. Base with wood screw. Projection 2-1/2 inches minimum. Diameter of base 1 inch.
Type F1323. - Cast iron or zinc alloy die casting. Otherwise the same as type 1323.
Type 1323AL. - Aluminum. Otherwise the same as type 1323.
Type 1324. - Flexible; wrought steel. Bright brass finish. 3 inch projection.
Type 1324B. - Same as type 1324 except dull bronze finish.
Type 1324L. - Same as type 1324 excpet 4 inch projection.
Type 1324BL. - Same as type 1324B except 4 inch projection.
Type 1325E. - Cast bronze. Diameter 4 inches. Convex rubber pad inserted from back into flanged rosette. Rosette secured to steel backplate attached to wall by expansion bolts.
Type 1325T. - Attachment to wall by toggle bolt. Otherwise the same as type 1325E.
Type 1326. - Cast bronze. Diameter minimum 2-1/4 inches. Convex or concave as required, rubber pad inserted from back to flanged rosette and attached to wall by toggle bolt. Rosette secured to backplate by three machine screws.
Type 1327. - Same as 1326 except rubber pad attached to rosette by concealed screws and backplat attached to wall by one wood screw.
Type 1327T. - Same as type 1327 except backplate attached to wall by toggle bolt.

3.7.26 Floor stops.

Type 1328. - Cast bronze, heavy weight. Rubber tip. Height 3 inches. Diameter of base 2-1/2 inches. Attachment to floor by three wood screws.
Type 1328E. - Same as type 1328, except for attachment to floor by steel stud with expansion shield and auxiliary machine screw with expansion shield.
Type 1329. - Same as type 1328, except with hook and keeper.

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Type 1329E. - Same as type 1328, except with hook and keeper, and attachment to floor by steel stud with expansion shield and auxiliary machine screw with expansion shield.
Type 1330. - For doors without thereshold. Cast bronze. Dome portion substantially quarter spherical. Molded rubber cushion fitting cavity in dome. Cushion 1-1/2 inches wide, 3/4 inch high and extending 1/4 inch beyond vertical face of dome. Cushion secured within dome by concealed fastenings. Rear portion of base shall have a 1/4 minimum diameter stud minimum 1/2 inch long projecting into floor to prevent rotation. Minimum thickness of base 5/32 inch. For attachment to wood floor by No. 14 wood screw.
Type 1330E. - For attachment to other than wood floors by 1/4-20 machine screw with expansion shield. Otherwise the same as type 1330.
Type 1330A. - For doors with threshold. Minimum thickness of base 1/2 inch. For attachment to wood floor. Otherwise the same as type 1330.
Type 1330AE. - For doors with threshold. Minimum thickness of base 1/2 inch. For attachment to other than wood floors by 1/4-20 machine screw with expansion shield. Otherwise the same as type 1330.
Type 1331. - Cast bronze. Rubber tip. Diameter of base 1-1/4 inches. Height 1-5/16 inches. Two wood screws.
Type F1331. - Cast iron or zinc alloy die casting. Otherwise the same as type 1331.
Type 1331AL. - Aluminum. Otherwise the same as type 1331.

3.7.27 Door silencers.

Type 1337A. - For metal frame. Molded rubber with underside of head recessed to form pneumatic chamber to cushion impact of door. Diameter or width 3/8".
Type 1337B. - For wood frames. Bronze plate if plate is used. Otherwise same as type 1337A.

3.7.28 Door roller bumpers. (For protecting interfering doors against striking each other.)

Type 1341. - Cast bronze with rubber roller vulcanized to a bronze bushing. Base approximately 2" in diameter. Project 3-3/8" minimum. Gray or white roller, not less than 1" in diameter by 3/8" thick.

3.7.2919 Hinged hasps.

Type F1401. - Wrought steel; heavy zinc coating. Brass pin. Dimensions as follows:

TABLE III. Dimensions for type F1401

Length closed	Size of screws		Minimum thickness of metal	
	Length	No.	Hasp	Staple plate
Inches	Inch		Inch	Inch
3	3/4	6	0.062	0.083
4-1/2	3/4	6	.074	.083
6	1	9	.084	.095
8	1	10	.096	.125

Type F1401C. - Wrought steel, zinc-plated. Steel pin. Otherwise the same as type F1401.
Type F1401E. - Wrought steel, unfinished. Steel pin. Otherwise the same as type F1401.
Type F1403. - Extra heavy. Wrought steel; heavy zinc coating. Brass pin. Length open, 7-1/2 inches; width, 1-3/4 inches; thickness, 0.0160 inch minimum. One square hole shall be provided, for fastening by carriage bolt if desired. (Carriage bolt is not to be furnished.)
Type F1403E. - Wrought steel, unfinished. Steel pin. Otherwise the same as type F1403.

3.7.29.1 Safety pattern types.

Type F1405. - Wrought steel; heavy zinc coating. Brass pin. Dimensions as follows: (See table below).

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TABLE IV. Dimensions for type F1405

Length closed	Length of joint	Size of screws		No. of screw holes	Minimum thickness of metal	
		Length	No.		Hasp	Staple plate
Inches	Inches	Inches			Inch	Inch
3-1/2	1-1/2	3/4	7	7	0.074	0.083
4-1/2	1-1/2	7/8	7	7	.086	.083
6	1-7/8	1	9	9	.104	.125
7	2-1/2	1-1/4	10	9	.156	.156

Type F1405C. - Wrought steel, zinc-plated. Steel pin. Otherwise the same as type F1405.

Type F1405E. - Wrought steel, unfinished. Steel pin. Otherwise the same as type F1405.

Type F1407. - Wrought steel, unfinished, steel pin. With adjustable staple. Dimensions as follows:

TABLE V. Dimensions for type F1407

Length closed	Length of joint	Size of screws		Minimum thickness of metal	
		Length	No.	Hasp	Staple plate
Inches	Inches	Inches		Inch	Inch
3-1/2	1-1/2	3/4	7	0.077	0.065
4-1/2	1-1/2	3/4	7	.089	.065
6	1-7/8	1	9	.107	.071

3.7.30 Hooks and eyes.

Type 1601. - Brass wire hooks and eyes. Hook lengths 1, 2, 3, 4, and 6 inches.

Type F1601. - Bright steel wire hooks and eyes. Hook lengths 1, 2, 3, 4, and 6 inches. Not plated.

3.7.31 Screen hangers.

3.7.31.1 For full length screens.

Type F1824. - Wrought steel, zinc-plated. Eye plate 2 inches x 1-7/8 inches. Hook plate 1-7/8 inches x 3/4 inch. Minimum thickness of metal 0.058 inch.

3.7.31.2 For half-length screens.

Type 1826. - Wrought steel, zinc-plated. Eye plate 1-5/8 inches x 1-1/4 inches. Hook plate 1-1/8 inches x 9/16 inch. Minimum thickness of metal 0.058 inch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. All hardware of the same type and size offered for delivery at one time shall be considered a lot for purposes of inspection and acceptance.

4.3 Sampling.

4.3.1 Sampling for visual and dimensional examination. Samples for visual and dimensional examination shall be selected in accordance with level S-1 of MIL-STD-105, with an AQL of 6.5 defects per hundred units.

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4.3.2 Sampling for acceptance testing. Samples for use in acceptance testing shall be selected in accordance with level S-1 of MIL-STD-105, with an AQL of 2.5 defects per hundred units.

4.4 Visual and dimensional examination. Visual and dimensional examination shall be made on items furnished under this specification as needed to insure conformance with the requirements for the specific types of items furnished.

4.5 Testing.

4.5.1 Tests for eletroplated coatings.

4.5.1.1 General.

4.5.1.1.1 Methods of Test. Electroplated coatings shall be tested by any of the applicable methods shown below for determining thickness. The salt spray test is for determining continuity of the coatings. At least three specimens of significant surfaces shall be utilized for each of the applicable tests for electroplated coatings.

Coating	Test Method	Test	Paragraph
Chromium plating on nickel	Spot test.....		4.5.1.2
Chromium plating on iron, steel and zinc alloy die casting	Salt spray test.....	ASTM B368	4.5.1.6
Nickel plating on iron or steel	Magnetic test.....	ASTM B530	4.5.1.3
	Microscopic test....	ASTM B487	4.5.1.4
Nickel plating on zinc alloy die casting or aluminum	Microscopic test....	ASTM B487	4.5.1.4
Cadmium plating on iron or steel	Coulometric test....	ASTM B504	4.5.1.5
	Magnetic test.....	ASTM B499	4.5.1.3
	Microscopic test....	ASTM B487	4.5.1.4
Zinc coating 1/ on iron or steel	Coulometric test....	ASTM B504	4.5.1.5
	Magnetic test.....	ASTM B499	4.5.1.3
	Microscopic test....	ASTM B487	4.5.1.4

1/ The tests apply to zinc coatings applied either by electroplating or hot dipping.

4.5.1.1.2 Umpire test. In case of dispute, the minimum thickness of nickel, copper, cadmium, and zinc coatings shall be determined by the microscopic test described in 4.5.1.4.

4.5.1.1.3 Significant surfaces for tests of minimum thickness of electroplated coatings. In general, "significant surfaces" are those surfaces that are visible and are subjected to wear or corrosion. For this specification they are defined as those visible areas that can be touched by the surface of a sphere 0.75 inch in diameter. For delineating these areas, a brass or steel sphere 0.75 ± 0.01 inch in diameter is attached to a small rod that serves as a handle. The surface of the sphere is coated with a thin film of colored wax. When the sphere is rubbed on the surface of the fitting, areas that cannot be marked are not significant and are not subject to test.

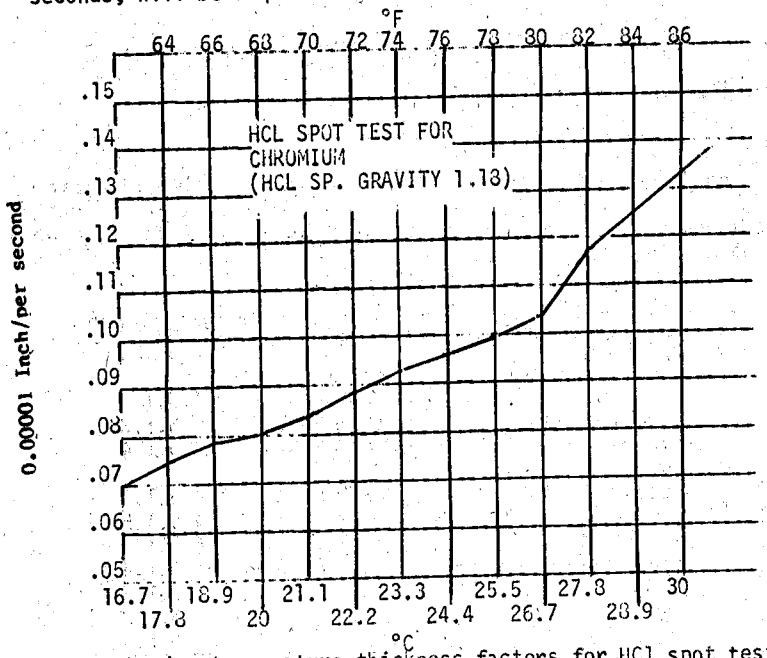
4.5.1.2 Spot test for minimum thickness of chromium plating on nickel.

4.5.1.2.1 Preparation. The spot test shall be applied at points on significant surfaces where the minimum thickness of chromium may be expected. At each spot to be tested, the surface shall be cleaned by light rubbing with a paste of pure magnesium oxide and water, followed by rinsing and drying. At each spot an area about 0.25 inch (6mm) in diameter shall be surrounded with a small ring of wax, or with a mark from a wax pencil. The article shall then be allowed to stand long enough (at least 15 minutes) to reach the room temperature which shall be between 65 and 85°F. (18 and 30°C.) and shall be noted within 2°F. or 1°C.

4.5.1.2.2 Procedure. Reagent quality hydrochloric acid, having a specific gravity at 60/60°F. of 1.180 ± 0.002 shall be used. (This corresponds to 11.5N ± 0.2N which may be checked by titration.) The acid shall also be kept in the room until it acquires room temperature. One drop (0.03 to 0.05 ml.) of the hydrochloric acid shall be placed within the wax ring with a dropper. The time required from the beginning of gas evolution to the first visibility of nickel shall be determined to the nearest second with a stop watch. If the bubbling does not start at once, the surface of the chromium within the ring may be touched with a fine nickel wire to start the action.

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4.5.1.2.3 Calculation of results. The thickness of the chromium at the spot tested shall be calculated from the accompanying curve by multiplying the factor for the temperature used in the test by the number of seconds required. For example, if 6 seconds are required at 72°F. (22°C.), reference to the curve gives a factor for this temperature of 0.036 ± 1.4 hundred-thousandths inch, or 0.000014 inch. To meet the specified minimum thickness of 0.00001 inch of chromium at any definite temperature, the following periods of time, in seconds, will be required at the temperature indicated, with 1.180 hydrochloric acid:



Curve showing temperature-thickness factors for HCl spot test for chromium plating (see 4.5.1.2.3).

Temperature		Seconds
F	C	
64	17.8	13.5
66	18.9	13
68	20	12.5
70	21.1	12
72	22.2	11.5
74	23.4	11
76	24.5	10.5
78	25.5	10
80	26.6	9.5
82	27.8	8.5
84	28.9	8

4.5.1.3 Magnetic test for minimum thickness of nickel, cadmium and zinc coatings. The significant surface specimens of nickel coated metals shall be tested in accordance with the applicable portions of ASTM B530 (see 3.2.6). The significiant surface specimens of cadmium and zinc coated metals shall be tested in accordance with the applicable portions of ASTM B499 (see 3.2.8 and 3.2.9, as applicable).

4.5.1.3.1 Applicability of magnetic test. The magnetic test is regularly used only for single coatings of substantially pure metal on brass, bronze, iron, and steel. It is not normally applicable to nickel plating over copper on steel, or to nickel coatings containing 10 percent or more of cobalt when applied to brass or bronze. For such coatings, the microscopic test (par. 4.5.1.4) shall be used. Any coating of chromium over the nickel coating on brass and bronze should be entirely removed. The methods given below (par. 4.5.1.3.2) shall be used for removing chromium and for determining the presence of sufficient cobalt in the nickel coating to affect the magnetic test. In tests of nickel coatings on iron and steel, the effect of chromium and cobalt is relatively insignificant, and the above precautions ordinarily are not necessary.

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4.5.1.3.2 Preparation of nickel coatings on brass and bronze for magnetic test. If chromium plated, completely immerse the piece in a solution containing about 10 percent by volume of chemically pure concentrated hydrochloric acid, which has been warmed to about 50°C. (122°F.). When all chromium has been dissolved it will be indicated by the ceasing of effervescence (usually in about 5 minutes). The piece shall then be heated in an air oven at 400°C. (752°F.) for 30 minutes and allowed to cool. This causes a yellowish tarnish of nickel coatings and a bluish tarnish of nickel deposits containing 10 percent or more of cobalt.

4.5.1.4 Microscopic test for minimum thickness of nickel, cadmium, and zinc coatings. The specimens of significant surfaces shall be tested in accordance with the applicable portions of ASTM B487 (see 3.2.6, 3.2.8 and 3.2.9, as applicable).

4.5.1.5 Test for minimum thickness of cadmium and zinc on iron and steel by the coulometric method. The specimens of significant surfaces shall be tested in accordance with the applicable portions of ASTM B504 (see 3.2.8 and 3.2.9, as applicable).

4.5.1.6 Salt spray test for continuity of chromium coatings on iron, steel and zinc alloy. The specimens of significant surfaces shall be tested in accordance with the applicable portions of ASTM B368 (see 3.2.5). The exposure period for the test shall be 16 hours.

4.5.2 Tests for plastics.

4.5.2.1 Test for deformation in hot water. Pieces for test shall be conditioned for at least 48 hours at 77° ± 2°F. and 50 ± 2 percent relative humidity, and shall then be immersed for 30 minutes in water kept at 160° ± 2°F. The pieces shall be suspended in the water by a cord or wire. Immediately after the 30 minute immersion, the items shall be examined for warping, bending, or contraction due to unmolding.

4.5.2.2 Accelerated aging test. Pieces for test shall be heated 48 hours at a temperature of 160°F. in a circulating air oven. After cooling to room temperature the items shall be examined for changes in shape, color, appearance and other physical characteristics readily determined by handling for inspection.

4.5.2.3 Test for flammability. Three specimens shall be tested in accordance with Federal Standard 406/6031.

4.5.2.5 Test for impact strength. Ten specimens for test shall be taken from the plate in both lengthwise and crosswise directions, tested singly, in edgewise position, and in accordance with Federal Standard 406/1071.

4.5.2.6 Boiling test. Three specimens 1 inch x 3 inches shall be cut from the plate and dried by heating at 50°C. for 24 hours. The weight of each specimen and its thickness at the central point shall be accurately determined. After boiling in water 2 hours, the weight and thickness shall be again determined immediately upon removal from the boiling water and quickly drying the surfaces. The change in weight and thickness shall be calculated to the nearest percent of the original dry weight and thickness. The samples shall then be examined for delamination, cracking, or splitting and for blistering, crazing, or checking of the surfaces.

4.5.2.7 Test for abrasion resistance. Three specimens shall be tested in accordance with Federal Standard 406/1091, Abrasion Wear (Loss in Weight). For each specimen, the loss in weight shall be determined for each 1,000 revolutions for a total of 5,000 revolutions.

4.5.2.8 Test for resistance to chemicals. Each chemical specified below shall be applied to the face of the plates in one or more spots approximately 1 inch in diameter, and the spots shall be covered with a petri dish. After 24 hours at room temperature (approximately 77°F.), the spot shall be washed, dried, and polished lightly with a soft, dry cloth. Careful examination shall be made for spotting and blistering. The chemicals shall be (a) trisodium phosphate, 5 percent solution; (b) phenol, 5 percent solution; and (c) carbon tetrachloride.

4.5.3 Tests for wood-bushed sash pulleys.

4.5.3.1 Field test. Sample pulleys shall be selected by the inspector and the wood bushings exposed. Upon heating bushing with small flame, the impregnating material shall melt, exude from wood and collect on surface. Upon removal of flame, exuded material shall solidify before cooling to room temperature and at room temperature shall not be sticky, gummy, or capable of being absorbed by ordinarly white blotting paper.

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4.5.3.2 Laboratory test. At the option of the inspector, wood bushings may be subjected to the following test. Bushings selected by the inspector shall be removed from the pulleys and accurately weighed. They shall then be immersed for not less than 12 hours in clean water at room temperature. Upon removal from water and immediate reweighing after quickly drying surface with absorbent material, the weight shall be not more than 5 percent greater than the original weight. The bushings shall then be placed in a desiccator containing concentrated sulphuric acid and dried for not less than 12 hours at room temperature, after which the weight shall be not more than 5 percent under the original weight. Bushings shall also withstand being maintained at a temperature of 120°F. to 125°F. with no indication of it becoming gummy or sticky. Upon heating above its melting point, the impregnating material shall exude free from wood and not be gummy or sticky.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, packaging, and marking. Preservation, packaging, packaging, and marking shall be in accordance with PPP-H-1581. Preservation and packaging shall be levels A or C, as specified (see 6.2). Packing shall be levels A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. Hardware covered by this specification is intended for use in shelving, cabinetry, and miscellaneous applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type and, if applicable, size of hardware desired (see 1.2.1).
(c) If a particular composition of zinc-castings is required (see 3.1.1.5).
(d) Standard finish required (see 3.2.1.1).
(e) If rust discoloration is not acceptable on iron or steel products (see 3.2.1.2).
(f) If particular hardware finishes are to be supplied without lacquer (see 3.2.2).
(g) Whether polished or unpolished finishes are desired (see 3.2.4).
(h) If hot-dip finish US 2H is required (see 3.2.9).
(i) Type of bolt with which strike is to be used (see 3.7.3 - Type 1048).
(j) If other than 12-foot standard with one style "A" rest per foot is required (see 3.7.8 - Type 1067).
(k) If more than one bracket for each 12 inch length of shelf standard is required (see 3.7.7.3).
(l) If door holder to accommodate other than 95° door opening (between 90° and 110°) is required (see 3.7.14.2.2 and 3.7.14.2.4).
(m) If 15 inch extension handle is required (see 3.7.14.2.3).
(n) If for doors with floor clearance greater than 1 1/2 inches (see 3.7.14.2.8).
(o) Whether wood screws, metal screws, or machine screws with expansion sleeves are required (see 3.7.14.2.8 - Type 1168).
(p) If beveled edges are required (see 3.7.18.1 - Types 1224, 1225, and 1226).
(q) Height and width required for door protection plates (see 3.7.18.1 - Types 1224, 1225, 1226, and 1227).
(r) If chocolate brown color is to be furnished (see 3.7.18.1 - Type 1227).
(s) Length of bar required (see 3.7.18.4).
(t) If bases are required with push-pull bar (see 3.7.18.4 - Type B1229S).
(u) If 2 1/2 inch diameter wheel is required (see 3.7.19 - Type 1230).
(v) If other than 6 foot length is required (see 3.7.21).

6.3 Types and finishes for "special application". Types designated herein by the letters "SA" prefixed to the type number are intended for certain special applications, and may not specified; when they are needed, the probable delivery date should be obtained from the source of supply before ordering.

6.4 Finishes.

6.4.1 Preferred finishes. As far as practicable, finishes should be confined to brass or satin bronze (solid or plated), nickel plated, chromium-plated, zinc-coated, and japanned. Zinc coating, of the thickness specified herein (3.2.9) is the preferred finish for steel intended for outdoor use.

6.4.2 Selection of finishes. Where finishes other than those recommended above are required, selection should be confined to finishes shown in table I.

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6.4.3 Standard finishes. Owing to differences in the processes of producing hardware finishes and the variety of metals to which they are applied, it is commercially impracticable to attain an exact match. Therefore, it is understood that hardware delivered in a standard finish from two or more sources will compare reasonably when the items are viewed at arm's length and approximately two feet apart. This approximates the conditions of use, as for example, different pieces of door hardware at various locations on the door. Samples of the standard finishes, at the time this specification was prepared, were furnished by the Builders' Hardware Manufacturers' Association 60 East 42nd Street, New York, N.Y. 10017.

TABLE VI. Standard finishes for builders' hardware
(see 6.4.3)

Symbol	General description	Application (Metal or surface)
USP	Primed for painting	----------
US1B	Bright japanned	----------
US1D	Dead black	----------
US2C	Cadmium plated*	Iron and steel
US2G	Zinc coated	Iron and steel
US2H	Zinc coated - hot-dipped	Iron and steel
US3	Bright brass	Brass, iron, and steel
US4	Dull brass	Brass, iron, and steel
US10	Dull bronze	Bronze, iron, and steel
US14	Nickel plated	Steel, polished surfaces
US26	Chromium plated	Bright or polished surfaces
US26D	Chromium plated, dull	Plain surfaces
US27	Satin aluminum, lacquered	Aluminum (except hinges)
US28	Satin aluminum, anodized**	Aluminum
US32	Stainless steel polished	Stainless steel
US32D	Stainless steel dull	Stainless steel

*Either cadmium or zinc, electroplated, may be furnished for US2C.
**Finish US28 is subject to slight variations in color when applied to different aluminum alloys. Ordinarily, such variations are not objectionable and are acceptable under this specification.

6.5 Dimensions. Where more than one dimension is given for items of builders' hardware, the vertical dimension is given first.

6.6 Definitions.

6.6.1 Definition of "builders' hardware". The term "builders' hardware" is intended to cover mechanical devices for the convenience, protection, and safety of the occupants of buildings; and for supporting, guarding, operating, controlling or securing the various movable parts of buildings, such as doors, windows, drawers, gates and scuttles.

6.6.2 Definition of "projection". For purpose of this specification "projection" means the distance that the complete item extends outward from the surface on which it is mounted.

6.7 New and deleted type numbers. Because of changes in demand for certain items of builders' hardware, new items have been added to this specification and other items present in the previous issues, FF-H-111A dated July 8, 1948 and FF-H-00111B dated June 10, 1960 have been deleted. The added and deleted items are listed by type number. Furthermore, some modifications have been made regarding size and finish requirements in some remaining types, and should be carefully noted.

TABLE VII. Cross-reference of hardware types

FF-H-111A	FF-H-00111B	FF-H-111C
1011	1011	1011
F1011	F1011	F1011
F1011C	F1011C	F1011C
1012	1012	1012
F1012	F1012	F1012
F1012C	F1012C	F1012C
1013	1013	1013
1021A	1021A	1021A
F1021A	F1021A	F1021A
1021B	1021B	1021B
F1021B	F1021B	F1021B

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TABLE VII. Cross-reference of hardware types (con.)

FF-H-111A	FF-H-00111B	FF-H-111C
1023A	1023A	1023A
F1023A	F1023A	F1023A
F1023AC	F1023AC	F1023AC
1023B	1023B	1023B
F1023B	F1023B	F1023B
F1023BC	F1023BC	F1023BC
1028A	1028A	Not Included
1028B	1028B	Not Included
SA1028C	SA1028C	Not Included
F1029	F1029	Not Included
1045	1045	1045
1045R	1045R	1045R
1046	1046	1046
1048	1048	1048
Not Included	1049	1049
Not Included	Not Included	1050AB
Not Included	Not Included	1050AS
F1051	F1051	F1051
1053	1053	1053
Not Included	1054	Not Included
1061	1061	1061
F1061	F1061	F1061
1062	1062	1062
F1062	F1062	F1062
1064	1064	Not Included
F1064	F1064	Not Included
Not Included	1064AL	Not Included
1065A	1065A	Not Included
Not Included	1065A-AL	Not Included
F1065A	F1065A	Not Included
Not Included	1065B	Not Included
Not Included	1065B-AL	Not Included
F1065A	F1065A	Not Included
F1065B	F1065B	Not Included
1065T	1065T	Not Included
Not Included	1065T-AL	Not Included
F1065T	F1065T	Not Included
F1066	F1066	F1066
F1066A	F1066A	F1066A
1067	1067	1067
F1067	F1067	F1067
1067G	1067G	1067G
F1067G	F1067G	F1067G
F1068	F1068	F1068
Not Included	Not Included	1068C
Not Included	Not Included	1068A
Not Included	Not Included	1068AA
Not Included	1069	Not Included
Not Included	F1069	Not Included
1070	1070	1070
1071	1071	1071
F1072	F1072	F1072
F1073	F1073	F1073
Not Included	F1073D	F1073D
Not Included	F1073S	F1073S
1074	Not Included	Not Included
F1074C	Not Included	Not Included
F1074D	Not Included	Not Included
1075	1075	1075
F1075	F1075	F1075
F1075A	F1075A	F1075A
F1075AC	Not Included	Not Included
1076	1076	Not Included
1076A	1076A	Not Included
F1076	F1076	Not Included
1077	Not Included	Not Included
F1077	Not Included	Not Included

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TABLE VII. Cross-reference of hardware types (con.)

FF-H-111A	FF-H-00111B	FF-H-111C
1078	Not Included	Not Included
1078A	Not Included	Not Included
F1078A	Not Included	Not Included
Not Included	1079	1079
1081	Not Included	Not Included
1082	Not Included	Not Included
1083	1083	1083
1085	1085	1085
F1085	F1085	F1085
Not Included	1086	1086
Not Included	F1086	F1086
1087	Not Included	Not Included
1088	1088	Not Included
1089	1089	1089
1090	1090	1090
Not Included	1090S	1090S
1091	1091	Not Included
1092	1092	Not Included
1096	1096	Not Included
F1096	Not Included	Not Included
1097	1097	Not Included
F1097	F1097	Not Included
1098	1098	Not Included
Not Included	1098AL	Not Included
1099	1099	Not Included
1101	1101	Not Included
1101A	1101A	Not Included
1102	Not Included	Not Included
1104	1104	Not Included
F1104	F1104	Not Included
F1105	F1105	Not Included
SA1106A	SA1106A	Not Included
SA1106AF	SA1106AF	Not Included
SA1106B	SA1106B	Not Included
SA1106BF	SA1106BF	Not Included
SA1106C	SA1106C	Not Included
SA1106CF	SA1106CF	Not Included
1108	1108	Not Included
F1108	F1108	Not Included
1108H	1108H	Not Included
F1108H	F1108H	Not Included
1112	Not Included	Not Included
F1112	F1112	Not Included
1114	Not Included	Not Included
F1114	Not Included	Not Included
1122	1122	Not Included
1123	1123	Not Included
F1123	F1123	Not Included
1125	1125	Not Included
1125A	Not Included	Not Included
1126	Not Included	Not Included
F1126	F1126	Not Included
SA1126	Not Included	Not Included
F1127	Not Included	Not Included
1128	1128	Not Included
F1128	F1128	Not Included
1128H	1128H	Not Included
F1128H	F1128H	Not Included
1129	1129	Not Included
F1129	F1129	Not Included
F1129A	Not Included	Not Included
1130	1130	Not Included
F1130	F1130	Not Included
Not Included	F1132	Not Included
F1133	F1133	F1133
1134	Not Included	Not Included

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TABLE VII. Cross-reference of hardware types (con.)

FF-H-111A	FF-H-00111B	FF-H-111C
1135	1135	1135
F1135	F1135	F1135
Not Included	1135AL	1135AL
1136	1136	1136
F1136	F1136	F1136
F1137	Not Included	Not Included
F1137C	F1137C	Not Included
1138	1138	Not Included
1139	1139	Not Included
Not Included	1139AL	Not Included
1140	1140	Not Included
F1140	F1140	Not Included
Not Included	1140AL	Not Included
1142	1142	Not Included
F1142	F1142	Not Included
1145	1145	Not Included
1146	1146	Not Included
F1147	Not Included	Not Included
F1147C	Not Included	Not Included
F1148	F1148	Not Included
F1148C	Not Included	Not Included
1149A	1149A	1149A
1149B	1149B	1149B
1154	1154	1154
1154A	Not Included	Not Included
Not Included	F1154	F1154
F1154A	Not Included	Not Included
1156	1156	1156
F1156	F1156	F1156
1158	1158	1158
1160	1160	1160
1160A	1160A	1160A
1160D	1160D	1160D
1161	1161	1161
1161A	1161A	1161A
1162	1162	1162
F1162	F1162	F1162
1164	1164	1164
1165	1165	1165
Not Included	Not Included	1165R
1166	1166	1166
1166A	1166A	1166A
1167	1167	1167
1168	1168	1168
Not Included	1168A	1168A
Not Included	1168B	Not Included
Not Included	1168C	Not Included
Not Included	1168D	Not Included
1169	1169	Not Included
F1170	F1170	Not Included
1171	1171	1171
F1171	F1171	Not Included
1172	1172	1172
F1172	F1172	F1172
Not Included	1172AL	1172AL
F1173	F1173	F1173
Not Included	Not Included	1174
Not Included	Not Included	1175
1176	Not Included	Not Included
1176A	Not Included	Not Included
F1188	F1188	Not Included
F1189	F1189	Not Included
F1190	F1190	Not Included
F1190C	Not Included	Not Included
F1195	F1195	Not Included
F1195C	F1195C	Not Included

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TABLE VII. Cross-reference of hardware types (con.)

FF-H-111A	FF-H-00111B	FF-H-111C
F1195H	Not Included	Not Included
1203	1203	1203
1204	1204	1204
F1204	F1204	F1204
F1204A	Not Included	Not Included
Not Included	1204AL	1204AL
1205	Not Included	Not Included
Not Included	1206	1206
1212	1212	1212
1213	1213	1213
F1213	F1213	F1213
Not Included	1213AL	1213AL
1223	1223	1223
F1223	Not Included	Not Included
F1223C	F1223C	F1223C
1224	1224	1224
1225	1225	1225
1226	1226	1226
1226A	1226A	1226A
1227	1227	1227
Not Included	Not Included	P1228
Not Included	Not Included	P1228P
Not Included	Not Included	PP1228W
Not Included	Not Included	B1228
Not Included	Not Included	B1228S
Not Included	Not Included	B1228PP
Not Included	Not Included	B1228PPS
Not Included	Not Included	B1228SD
1229F	Not Included	Not Included
1229FS	Not Included	Not Included
1229L	Not Included	Not Included
1229LS	Not Included	Not Included
1230	1230	1230
1230A	1230A	1230A
1230B	1230B	1230B
SA1231	SA1231	SA1231
SA1231A	SA1231A	SA1231A
1232	Not Included	Not Included
F1232	Not Included	Not Included
F1233	Not Included	Not Included
1236	1236	1236
1236A	1236A	1236A
1236C	1236C	1236C
F1240	F1240	Not Included
1240	1240	Not Included
F1243	F1243	Not Included
F1243A	F1243A	Not Included
F1245	F1245	Not Included
Not Included	1246A	Not Included
Not Included	F1246A	Not Included
Not Included	1247B	Not Included
Not Included	1248	Not Included
Not Included	1253	Not Included
Not Included	1254	Not Included
1262	1262	Not Included
1264	1264	1264
1265A	1265A	1265A
1265B	1265B	1265B
Not Included	1265AL	1265AL
1266	1266	1266
1269	1269	1269
F1269	F1269	F1269
1274	1274	1274
F1274	Not Included	Not Included
1275	1275	1275
F1275	Not Included	Not Included
F1275C	F1275C	F1275C

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TABLE VII. Cross-reference of hardware types (con.)

FF-H-111A	FF-H-00111B	FF-H-111C
F1276	F1276	F1276
F1276C	Not Included	Not Included
1280	1280	1280
1280D	1280D	1280D
1280P	Not Included	Not Included
1281	1281	1281
1282	1282	1282
1283	Not Included	Not Included
F1284	Not Included	Not Included
Not Included	1285	1285
Not Included	1286	1286
Not Included	1286AL	1286AL
Not Included	1287	1287
1305	1305	1305
1306	1306	1306
1306D	1306D	1306D
1306P	Not Included	Not Included
1307	1307	1307
F1307	F1307	F1307
1308	1308	1308
F1308	F1308	F1308
1310	1310	1310
F1310	F1310	F1310
F1315	F1315	Not Included
F1317	F1317	F1317
1320	1320	1320
1320E	1320E	1320E
1321	1321	1321
1321E	1321E	1321E
1322	1322	1322
F1322	F1322	F1322
Not Included	1322AL	1322AL
1323	1323	1323
F1323	F1323	F1323
Not Included	1323AL	1323AL
Not Included	Not Included	1324
Not Included	Not Included	1324B
Not Included	Not Included	1324L
Not Included	Not Included	1324BL
1325E	1325E	1325E
1325T	1325T	1325T
1326	1326	1326
1326T	1326T	Not Included
Not Included	1327	1327
Not Included	1327T	1327T
1328	1328	1328
1328E	1328E	1328E
1329	1329	1329
1329E	1329E	1329E
1330	1330	1330
Not Included	1330E	1330E
1330A	1330A	1330A
Not Included	1330AE	1330AE
1331	1331	1331
F1331	F1331	F1331
Not Included	1331AL	1331AL
1337A	1337A	1337A
1337B	1337B	1337B
1338	1338	Not Included
1341	1341	1341
1341A	1341A	Not Included
1342	1342	Not Included
1344	1344	Not Included
F1401	F1401	F1401
F1401C	F1401C	F1401C
F1401E	F1401E	F1401E

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TABLE VII. Cross-reference of hardware types (con.)

FF-H-111A	FF-H-00111B	FF-H-111C
F1403	F1403	F1403
F1403C	Not Included	Not Included
F1403E	F1403E	F1403E
F1405	F1405	F1405
F1405C	F1405C	F1405C
F1405E	F1405E	F1405E
Not Included	F1407	F1407
F1430	Not Included	Not Included
F1430C	F1430C	Not Included
F1430D	F1430D	Not Included
F1510	F1510	Not Included
F1520	F1520	Not Included
F1520A	F1520A	Not Included
F1530	F1530	Not Included
F1530A	F1530A	Not Included
1601	1601	1601
F1601	F1601	F1601
F1650	F1650	Not Included
F1651	Not Included	Not Included
F1651A	Not Included	Not Included
F1653	F1653	Not Included
F1654	Not Included	Not Included
F1655	F1655	Not Included
F1801	F1801	Not Included
F1801C	Not Included	Not Included
F1804	F1804	Not Included
F1804C	Not Included	Not Included
F1805	F1805	Not Included
F1805C	Not Included	Not Included
F1806	F1806	Not Included
F1806C	Not Included	Not Included
F1811	Not Included	Not Included
F1811C	Not Included	Not Included
F1812	Not Included	Not Included
F1812C	Not Included	Not Included
Not Included	1813	Not Included
F1821	F1821	Not Included
F1821A	F1821A	Not Included
Not Included	F1821AL	Not Included
Not Included	F1821A-AL	Not Included
F1824	F1824	F1824
F1825	Not Included	Not Included
F1825B	Not Included	Not Included
F1826	F1826	F1826
F1830	F1830	Not Included
F1831	F1831	Not Included
F1840	F1840	Not Included
F1845	F1845	Not Included
Not Included	1850	Not Included
Not Included	1850C	Not Included
Not Included	1855	Not Included
Not Included	1856	Not Included
Not Included	1857	Not Included
Not Included	1858	Not Included
Not Included	1860	Not Included
Not Included	F1860	Not Included
Not Included	1861	Not Included
Not Included	F1861	Not Included
Not Included	1862	Not Included
Not Included	1863	Not Included
Not Included	1864	Not Included
Not Included	1870	Not Included
Not Included	1871	Not Included
Not Included	1872	Not Included
Not Included	1873	Not Included
Not Included	1874	Not Included
Not Included	1875	Not Included

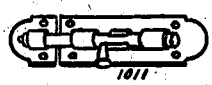
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TABLE VII. Cross-reference of hardware types (con.)

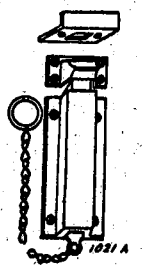
FF-H-111A	FF-H-90111B	FF-H-111C
Not Included	1876	Not Included
Not Included	1877	Not Included
Not Included	1878	Not Included
Not Included	1879	Not Included
Not Included	1880	Not Included
Not Included	1881	Not Included
Not Included	1882	Not Included
Not Included	1883	Not Included
Not Included	1884	Not Included
Not Included	1890	Not Included
Not Included	1890A	Not Included
Not Included	1891	Not Included
Not Included	1892	Not Included
Not Included	1893	Not Included
Not Included	1894	Not Included
Not Included	1895	Not Included
Not Included	1896	Not Included
Not Included	1897	Not Included
Not Included	1898	Not Included
Not Included	1899	Not Included
Not Included	1900	Not Included
Not Included	1901	Not Included
Not Included	1902	Not Included
Not Included	1905	Not Included
Not Included	F1905	Not Included
Not Included	1906	Not Included
Not Included	F1906	Not Included
Not Included	1907	Not Included
Not Included	1908	Not Included
Not Included	1909	Not Included
Not Included	1910	Not Included
Not Included	1911	Not Included
Not Included	1912	Not Included
Not Included	1912Z	Not Included
Not Included	1913	Not Included
Not Included	1913Z	Not Included
Not Included	1914	Not Included
Not Included	1915Z	Not Included
Not Included	1916	Not Included
Not Included	1917Z	Not Included
Not Included	1918	Not Included
Not Included	F1918	Not Included
Not Included	1919	Not Included
Not Included	1920	Not Included

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TYPE 1011



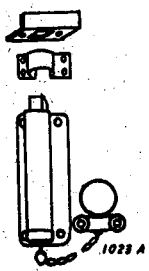
TYPE 1021A



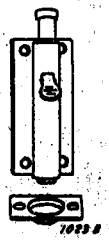
TYPE 1021B



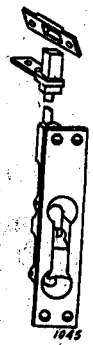
TYPE 1023A



TYPE 1023B



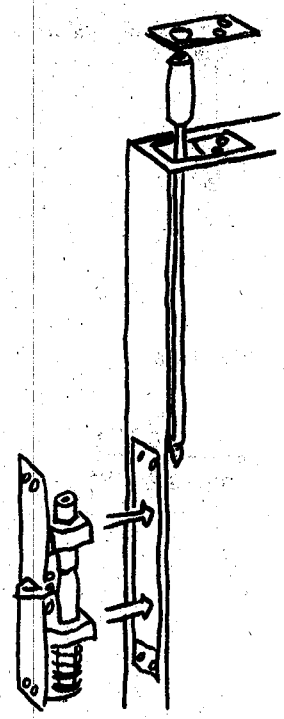
TYPE 1045



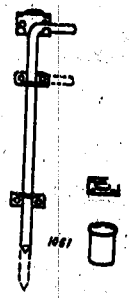
TYPE 1048



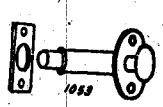
Type 1050AB



TYPE 1051

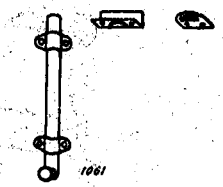


TYPE 1053



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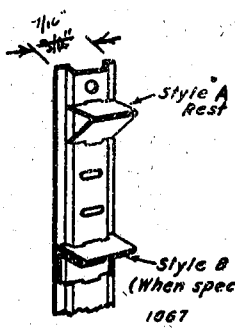
TYPE 1061



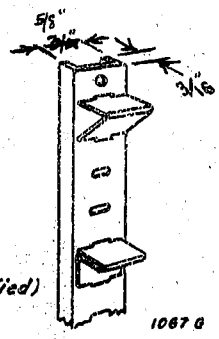
TYPE 1066



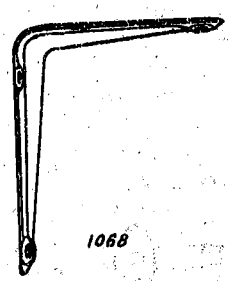
TYPE 1067



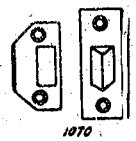
TYPE 1067G



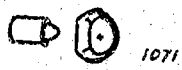
TYPE 1068



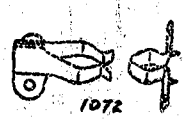
TYPE 1070



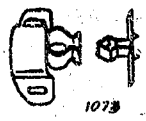
TYPE 1071



TYPE 1072



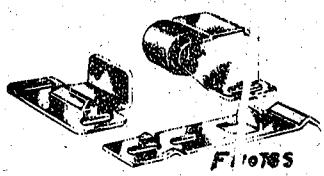
TYPE 1073



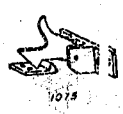
TYPE F1073D



TYPE F1073S



TYPE 1075

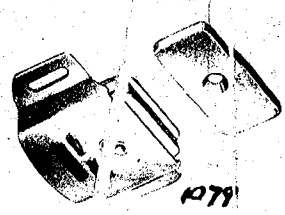


TYPE 1075A

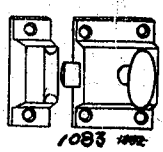


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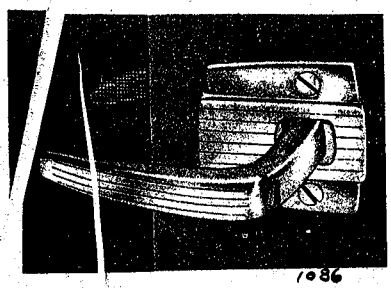
TYPE 1079



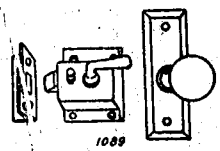
TYPE 1083



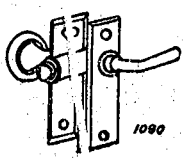
TYPE 1086



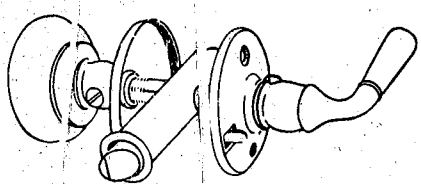
TYPE 1089



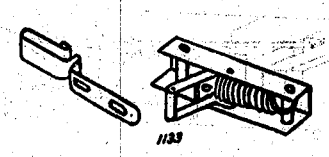
TYPE 1090



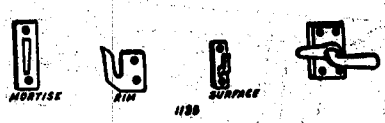
TYPE 1090S



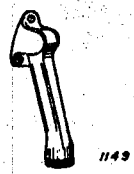
TYPE 1133



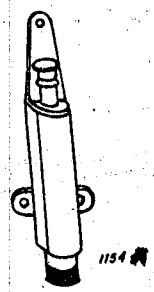
TYPE 1135



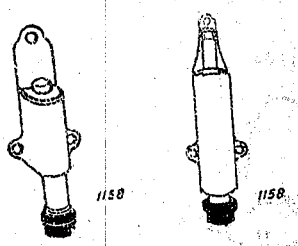
TYPE 1149



TYPE 1154A

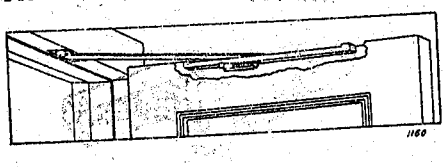


TYPE 1158

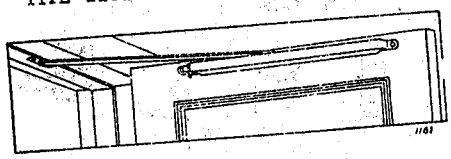


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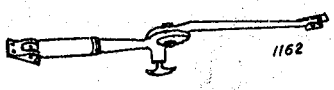
TYPE 1160



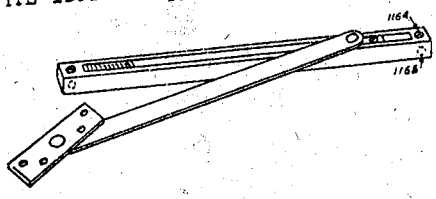
TYPE 1161



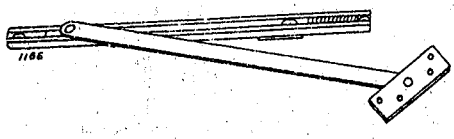
TYPE 1162



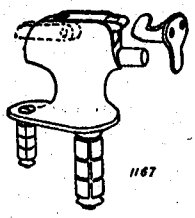
TYPE 1164 TYPE 1165



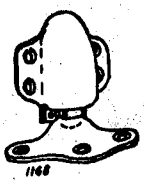
TYPE 1166



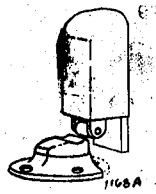
TYPE 1167



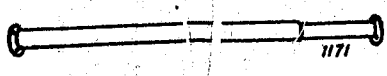
TYPE 1168



TYPE 1168A



TYPE 1171



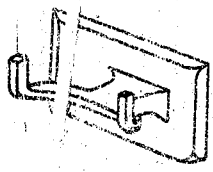
TYPE 1172



TYPE 1173

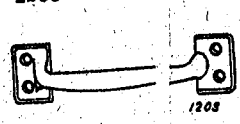


TYPE 1174



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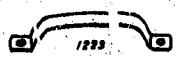
TYPE 1203



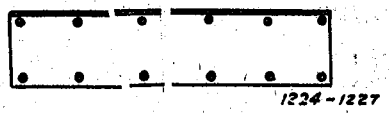
TYPE 1212



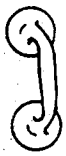
TYPE 1223



TYPE 1224 — TYPE 1227



TYPE P1228



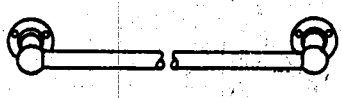
TYPE P1228P



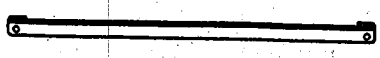
TYPE PP1228W



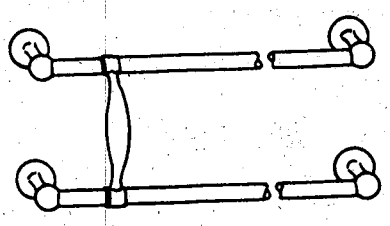
TYPE B1228



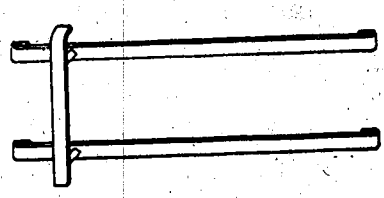
TYPE B1228S



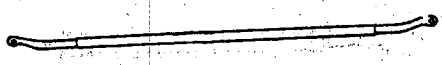
TYPE B1228PP



TYPE B1228PPS



TYPE B1228SD

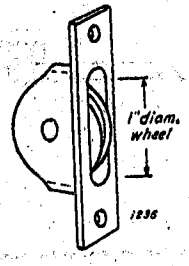


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TYPE 1230



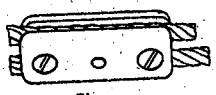
TYPE 1236



TYPE 1236A

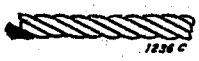


Soldering lug



Clamp

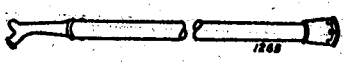
TYPE 1236C



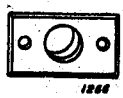
TYPE 1264



TYPE 1265

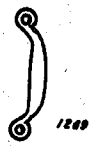


TYPE 1266





TYPE 1269



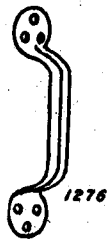
TYPE 1274



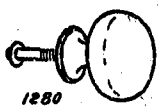
TYPE 1275



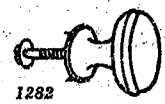
TYPE 1276



TYPE 1280



TYPE 1282



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TYPE 1285



TYPE 1286



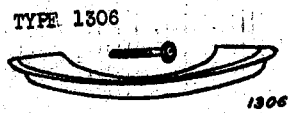
TYPE 1287



TYPE 1305



TYPE 1306





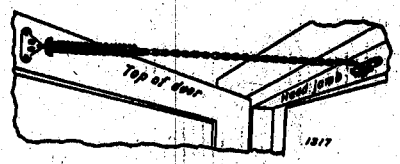
TYPE 1308



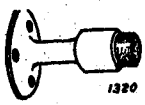
TYPE 1310



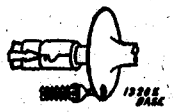
TYPE 1317



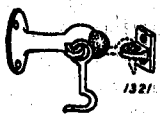
TYPE 1320



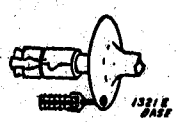
TYPE 1320E



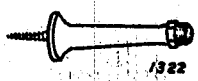
TYPE 1321



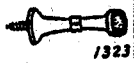
TYPE 1321E



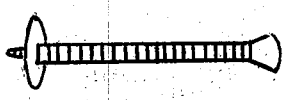
TYPE 1322



TYPE 1323



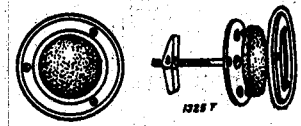
TYPE 1324



TYPE 1325E

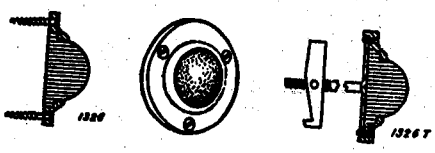


TYPE 1325T



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TYPE 1326 TYPE 1326T



TYPE 1328 TYPE 1328E



TYPE 1329 TYPE 1329E



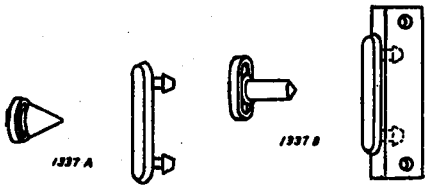
TYPE 1330



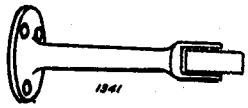
TYPE 1331



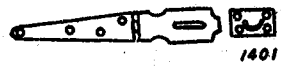
TYPE 1337A TYPE 1337B



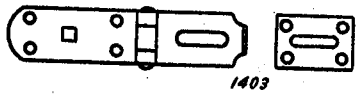
TYPE 1341



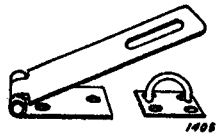
TYPE 1401



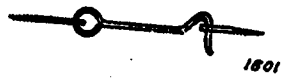
TYPE 1403



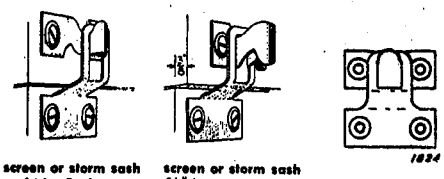
TYPE 1405



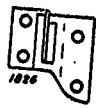
TYPE 1601



TYPE 1824



TYPE 1826



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